NOT MEASUREMENT SENSITIVE

A-A-58033
October 10, 1994

COMMERCIAL ITEM DESCRIPTION

REFLECTOMETER, METALLIC, TIME DOMAIN

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies

1. SCOPE

1.1 Scope. This Commercial Item Description (CID) describes a cable tester. This CID is meant as a minimum requirement for the cable tester in which only those manufacturers which meet or exceed the following requirements are qualified per this CID. This is a portable, self-contained, solid state, long range cable tester that uses time domain reflectometry to identify and locate cable faults. A signal output is required to interface the cable tester with an external printer or plotter.

2. CLASSIFICATION

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 3, style C, and color R in accordance with MIL-T-28800.

3. SALIENT CHARACTERISTICS

3.1 Performance Characteristics.

3.1.1 Vertical system. The vertical system shall be directly calibrated in decibels (dB). A front-panel adjustment shall enable the operator to establish a reference level by comparison with a precision 50 ohm +- 2% termination.

3.1.1 Dynamic range. The equipment shall have the capability to measure losses from 0 dB to at least 60 dB with an accuracy of +/-3% full scale.

3.1.1.2 Displayed noise. The displayed noise shall be less than -80 dB rms.

3.1.1.3 Noise suppression. A front-panel switch to permit low-noise operation shall be provided. When operated, the switch shall engage a noise filter that limits the displayed noise to less than -86 dB rms.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: The Resources and Logistics Services Division, SA-ALC/TILDD, Bldg. 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6524

AMSC N/A

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FSC 6625

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3 1 2 Horizontal system The horizontal system shall be directly calibrated in feet with a resolution of 0 04 feet (ft) or better

3 1 2 1 Measurement range The equipment shall have a measurement range of 0 to at least 5000 ft

3 1 2 2 Distance scale selector A switch shall be provided for feet/division (ft/div) control delineating ranges of 5, 10, 25, 100, 250, and 500 ft/div in the X10 mode and 50, 100, 250, 500, 1000, 2500, and 5000 ft/div in the X100 mode

3 1 2 3 Accuracy The accuracy shall be within +/-2% of reading plus the uncertainty of the propagation velocity factor

3 1 2 4 Propagation velocity factor A provision shall be made to compensate for different propagation velocity factors ranging from 0 30 to 0 99 increments of 0 1 via a front-panel adjustment and shall be accurate to within 1%

3 1 2 5 Sweep repetition The sweep repetition shall be automatic The rate shall be at least 40 Hz +/-20% with the noise filter energized

3 1 3 Display A cathode ray tube (CRT) or liquid crystal display (LCD) shall be provided for visual observation of the displayed signal The display shall have a minimum useful display area of 50 mm by 90 mm

3 1 3 1 Internal graticule An internal graticule with 8 vertical and 10 horizontal divisions shall be provided Subdivision markings of 0 2 division shall be provided along the vertical and horizontal center axes

3 1 3 2 Vertical position control A vertical position control shall be provided This control shall have a range sufficient to position the trace on any desired horizontal graticule of the display at all deflection factor settings

3 1 3 3 Zero reference set A horizontal position control shall be provided This control shall have a range sufficient to position the incident pulse edge to the vertical reference line of the display when the distance dial is at zero or when the zero reference switch is engaged

3 1 3 4 Zero reference A momentary contact switch shall be provided to check the horizontal reference of the incident pulse edge on the display

3 1 4 Recorder output Provisions shall be made to interface with an external standard commercial printer or plotter through either a GPIB RS-232 or X-Y interface

3 1 5 Input protection Protection shall be provided to allow for inputs of up to 400 volts (DC + peak AC up to 440 Hz)

3 1 6 Output impedance The tester cable shall provide multiple impedances of 50, 75, 93, and 125 ohms selectable by a front panel control

3 1 7 Test signal The equipment shall be capable of generating a half-sine wave excitation signal The signal amplitude shall be 5 volts + -10% unterminated 2 5 volts +/-10% terminated

3 1 7 1 Reflected signal The half sine wave signal shall have a rise time of less than 140 picosec when reflected from the 50 ohm accessory cable terminated into an open circuit

3 1 7 1 1 Signal aberrations The signal aberrations shall be no more than -30 dB peak-to-peak

3 1 7 1 2 Signal jitter The signal jitter shall be no more than 2 nsec in the X10 range setting and no more than 20 nsec in the X100 range setting

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3.1.8 Accessories The following accessories shall be provided with each unit and all accessories except a tester cable slide rule shall be stored in the front cover

(1) One 50 ohm precision termination
(2) One tester cable slide rule
(3) One replacement fuse
(4) One AC power cord
(5) Two operators manuals
(6) One precision 50 ohm cable 9 ft long
(7) One BNC female-to-female adapter

3.1.9 Physical characteristics

3.1.9.1 Tilt bail handle The equipment shall be supplied with a tilt handle The handle shall permit the equipment to be carried face up with the longest dimension vertical and shall tilt and support the chassis for easier viewing

3.1.9.2 Dimensions and weight The size and weight shall be consistent with current commercial capabilities and shall not exceed the maximum dimensions for shipboard applications The nominal overall size should be 13 in (330 mm) wide by 5 in (130 mm) high by 18 3/4 in (480 mm) deep with the tilt bail handle positioned to carry the equipment face up The weight should not exceed 25 lb (11 kg)

3.1.10 Electrical Power Sources The equipment shall operate from nominal commercial military and shipboard power sources of 110/220 VAC (+10%) at line frequencies of 45 Hz to 440 Hz Power consumption shall not exceed 25 watts Transient state conditions shall meet MIL-T-28800 requirements for the classification specified herein An internal DC source shall also be provided and shall have a battery operating time of at least 5 hours after a recharge time not to exceed 20 hours An internal charger shall also be provided

3.1.11 Temperature requirements

3.1.11.1 Operating temperature The equipment shall meet its performance and accuracy requirements in an operating environment of -10 deg C to +55 deg C

3.1.11.2 Nonoperating temperature The equipment shall meet its performance and accuracy requirements in a nonoperating environment of -20 deg C to +55 deg C

3.1.12 Mean time between failures (MTBF) The equipment MTBF shall be 2,000 hours The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours

3.1.13 Pre-conditioning requirement Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling with thermal cycling being accomplished in the sample testing phase

3.1.14 Calibration and maintenance adjustments The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments These adjustments shall be provided by variable value components which are adjustable by the use of simple means The calibration by substitution of selected components or parts is unacceptable unless specifically approved The calibration adjustments wherever possible shall be accessible without removal of the instrument case or modules The calibration interval shall be a period of 1 year or greater based on an operating time of 2,000 hours

4 REGULATORY REQUIREMENTS

4.1 Regulatory Requirement The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR)

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5 QUALITY ASSURANCE PROVISIONS

5.1 Quality Assurance Provisions

5.1.1 Contractor Certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings specifications standards and quality assurance practices The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

6 PACKAGING

6.1 Packaging Preservation, packing and marking shall be as specified in the contract or order

7 NOTES

7.1 Source of government documents Copies of Military and Federal documents are available from

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia PA 19111-5094

7.2 Products known to meet CID The following products are known to have previously met the requirements of this CID and need not resubmit bid samples or certification All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required

Manufacturer	Model	NSN
Tektronix	1503B w/opt 02	6625-01-283-2334
Tektronix	1503C w/opt 03	none

Custodian

Air Force - 99

Preparing Activity

Air Force - 82

Reviewing Activities

Navy - SH
DLA - ES

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